Manufacturing, Materials & Cost

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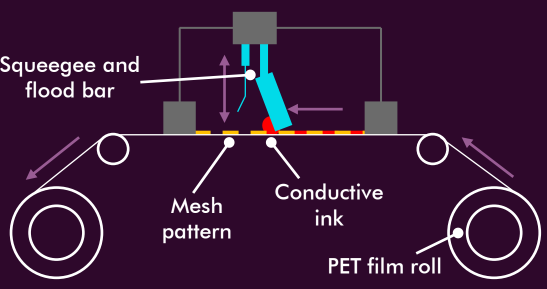
Dance Mat

Graphite on PET film

The circuit layers are transparent PET (Polyethylene Terephthalate) films with screen printed conductive graphite tracks.

PET is used here because it is **cheap**, **flexible**, **lightweight** and can be readily printed on in bulk. Conductivity is achieved by using a graphite ink in the roll-to-roll screen-printing process (below) to create a cost-effective method of printing conductive traces (compared to copper tracks for example).

Roll-to-roll screen printing



In the dance mat, graphite has been printed on a PET film in a roll-to-roll screen printing process to produce large (1m^2) flexible circuits.

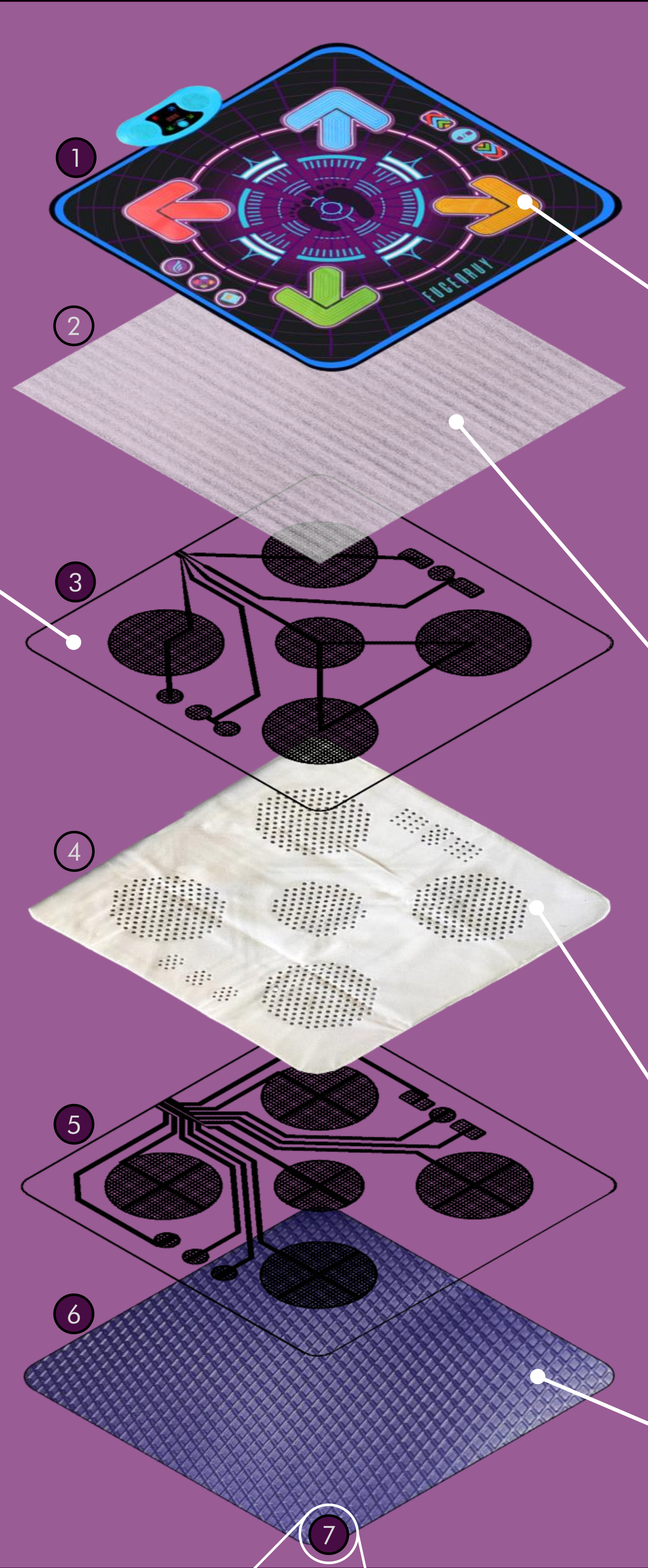
In this process the film is fed under a patterned mesh with gaps corresponding to the graphite tracks on the circuit layer. A flood bar then covers the screen with conductive graphite ink before the squeegee pushes the ink through the gaps in the mesh – on to the PET substrate.

This process is used for the flexible circuits not only because it is very fast and cost effective, but also because of the simplicity of the tooling and setup and crucially the ability to use conductive ink.

	Part N°	Description	Material	Qty	Cost estimation (£)
Dance Mat	1	Mat top	PEVA	1	0.40
	2	Foam cushioning layer	EPE	1	0.20
	3	Ground switch layer	PET/Graphite	1	1.00
	4	Foam spacer layer	EPE	1	0.20
	5	Live switch layer	PET/Graphite	1	1.00
	6	Mat bottom	PEVA	1	0.40
	7	Edge piping	Nylon	1	0.30
Controller	8	Housing top	ABS	1	0.40
	9	Housing bottom	ABS	1	0.25
	10	Display film	PET	1	0.10
	11	Speaker	Mixed	1	1.30
	12	Battery cover	ABS	1	0.10
	13	Battery connecting clips	Steel	3	0.25
	14	Control PCB	Mixed	1	2.00
	15	Volume knob	ABS	1	0.10
	16	M4 Self-tapping screws (silver)	Steel	2	0.05
	17	M3 Self-tapping screws (black)	Steel	10	0.25
				Total estimated materials cost:	**£8.30**

Plastic   Foam   Metal

ABS parts costed by weight at £0.86/kg [1]. EPE, PET, Nylon parts cost estimated per sheet from m^2 price for bulk listings on Alibaba.com [2]. Electronics and steel costed using components of very similar complexity/size on Alibaba.com [2].



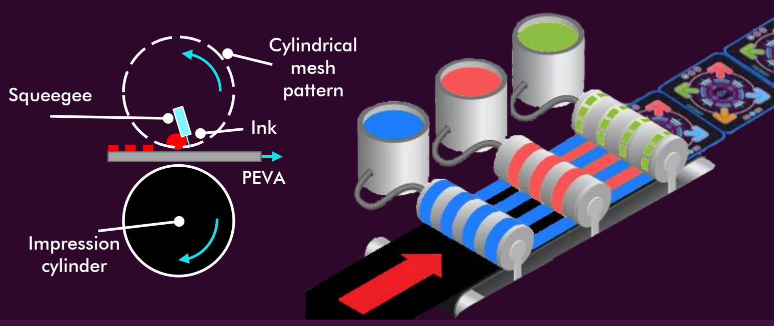
PEVA top surface

The top surface is made from PEVA (like a shower curtain). This is **water-resistant**, **wipeable** and **smooth** (all important when dealing with children!).

The top layer protects the mat from inevitable spillages and is easy to clean whilst also being tough enough to withstand vigorous dancing! It is smooth, but not so slippery that the user falls over while dancing.
PEVA is quite crinkly, and creases easily when folded repeatedly for storage of the dance mat.

The artwork has been rotary screen printed. This is much more cost effective for bulk production and produces a more durable print than digital printing.

Rotary screen printing



EPE foam padding

An EPE (expanded polyethylene) foam layer makes the mat **less crinkly** and **softer** to stand on. It also provides some protection to the printed circuit below to reduce wear/damage during use.

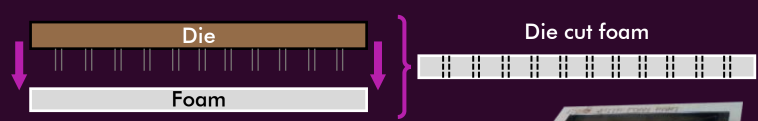
Pool noodles [3] and foam life jackets contain EPE for its water repellent and low-density qualities. In the mat these properties are used to provide another layer of protection for the circuitry underneath as well as some lightweight cushioning for the user's feet.

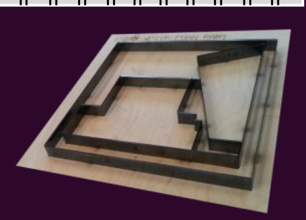
Foam spacer layer

A layer of low-density foam separates the two switch layers to provide mechanical resistance to the switches being pressed. It is die cut, likely in a large hydraulic press.

Altering the thickness and hole density of the spacer foam changes the force needed to press a switch. Therefore, this material has been selected (and the thickness and hole pattern designed) to ensure the activation force is not too heavy for children to press, but also so that switches cannot easily be activated by accident.

Hydraulic die cutting





A custom die with steel blades (similar to the one shown [4]) is a common method for producing bulk amounts of cut foam.

Often done with a hydraulic press, this process has a very low tooling cost, but if the design is changed then the whole die must be remade while production stops – a laser cut file for example could be changed immediately.

PEVA non-slip base

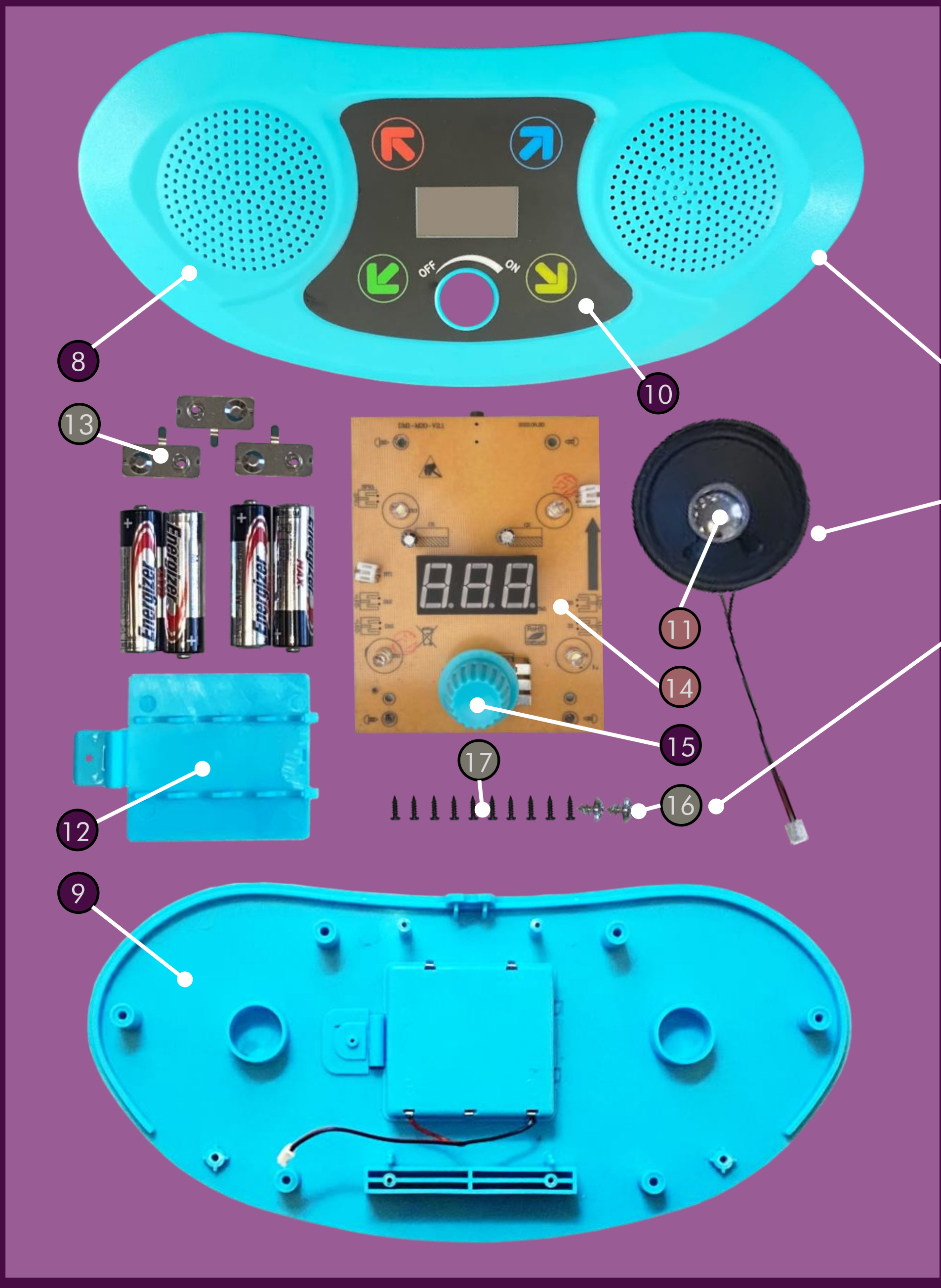
Like the top surface, the bottom surface is PEVA. Here it is textured with a diamond pattern to increase its friction with the floor, so it doesn't slip during use (as previously investigated).

The diamond pattern is imprinted on the PEVA by feeding the sheet between embossing rollers [5]. Adding a texture to PEVA like this is a cheap way to improve the grip of the mat on the floor without adding more expensive non-slip Materials like a rubber base.

Mat assembly - Nylon edge piping

The 6 layers of the mat are sewed together around the edge and covered with a 3.3m length strip of nylon fabric.

Each stitch is approximately 5mm long, meaning a sewing machine at 1500 stitches/min would be able to sew the mat in just under 30 seconds.
The stitched edge also keeps the layers of the mat from slipping against each other during use, so the user doesn't fall over!



Controller

Injection moulded parts

The upper and lower parts of the controller housing, as well as the battery cover and volume knob are injection moulded in cyan ABS. The ABS is 5mm thick in most places making the controller housing very sturdy and rigid. It can easily withstand being dropped or stood on (tested up to 80 kg!).

Electronics

The speaker and PCB are cheap – but relatively expensive in the controller and make up almost 70% of the estimated cost of the controller

Fasteners

10 black M3 x 10 and 2 silver M4 x 5 self-tapping screws are used to hold to 2 halves of the housing together and to fix the dance mat to the housing, respectively. Self-tapping screws speed up assembly time (no nuts!) and all screws can be done up with the same common Phillips head screwdriver

Controller assembly

After a few practice attempts the controller could be reliably assembled in 2:30s. A fast worker doing this full time could likely do it in under **2 minutes**. Using the length of the edge piping and taking average sewing machine speed as 1500 stitches/min gives a time of **30 seconds** to stitch the edge of the mat.

In total, each mat could be assembled in under 2:30s.
For a production run of 10,000 this results in 410 man-hours of labour, which for a group of 10 workers working 12-hour days would take under 4 days!

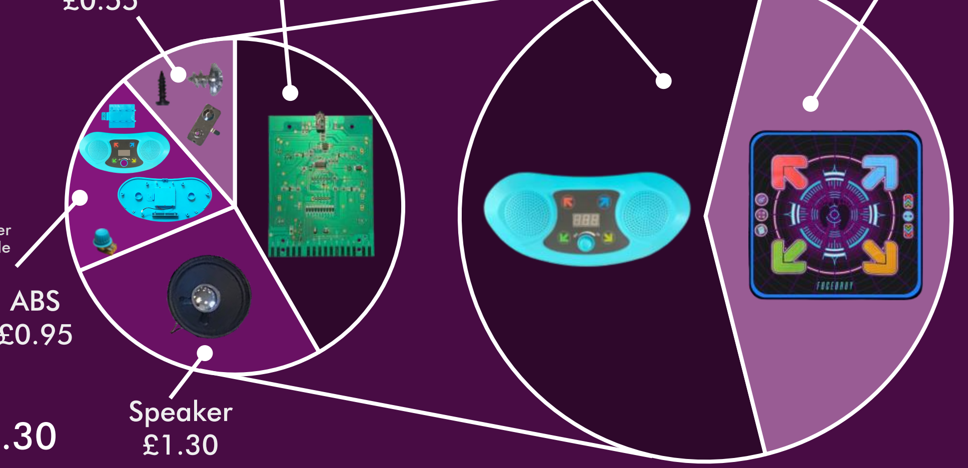
In reality, production would likely take more than a week, but this does highlight that the simplicity of the products makes assembly very quick – shortening delivery time and reducing labour costs.

Cost Analysis

The 2 subassemblies make up roughly equal parts of the material cost.

Within the dance mat the 2 intricate circuit layers represent the largest cost (60%), while the foam and PEVA layers are relatively minor costs.

In the controller, the electronics (speaker and PCB) are 70% of the total cost, while ABS parts and fasteners represents the minor costs.



Retail price: £44
Material cost: £8.30

Deep Dive: Mat Layer Manufacturing

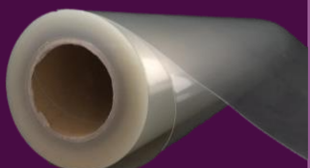
Flexible circuit layers

Material selection

PET (Polyethylene Terephthalate) film is used as the substrate for the printed circuit layers.

PET is available in bulk for 50p/metre and sheets are around 130 µm thick [2] – making it ideal for use in the dance mat.
Including the cost of screen printing, the cost of each circuit layer would probably not exceed £1 – an extremely cheap way to produce very large flexible circuits!

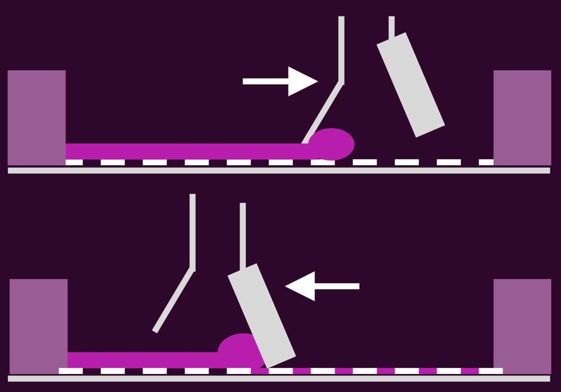
Also, the lightweight, thin film means there is no impact on the user feel or a reduction in softness, while also keeping the mat light and easy to move/store



The conductive traces on the film are printed graphite infused ink

Screen printing

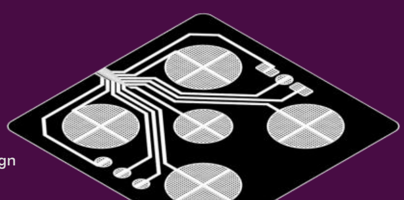
Roll-roll screen printing is used to print the conductive graphite tracks on to the PET substrate.
A more detailed look at the screen-printing process is shown below:



Firstly, the mesh is covered with a thin layer of ink. This is done using the flood bar which scrapes a thin layer of the graphite ink across the mesh.
Next, a squeegee runs the other way to force ink through the holes in the mesh, onto the PET substrate below.

This is process is continuously repeatedly as the rolled sheet of PET is fed under the mesh screen.

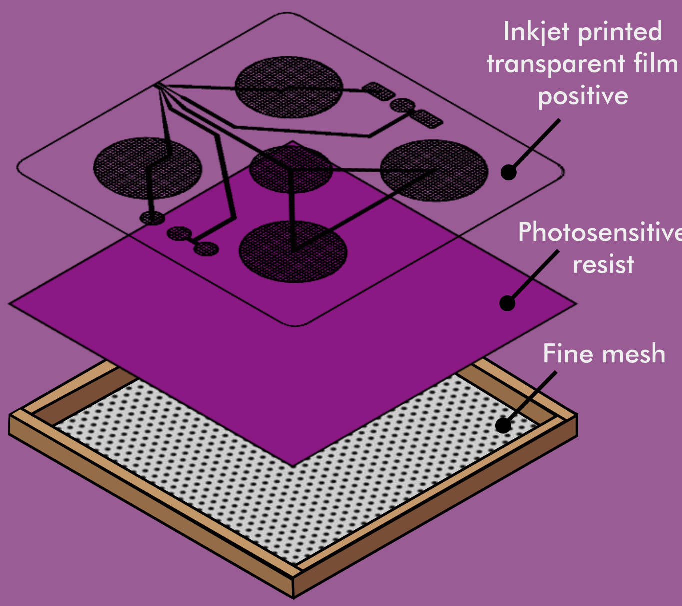
Resists & Patterns

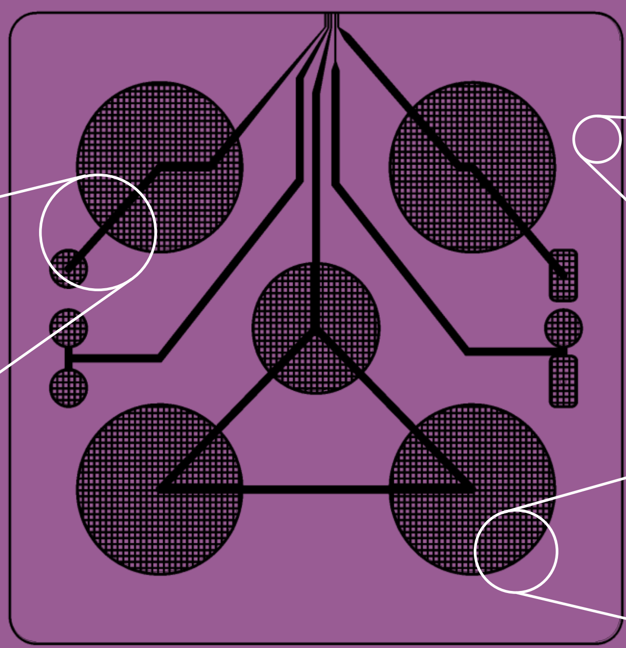


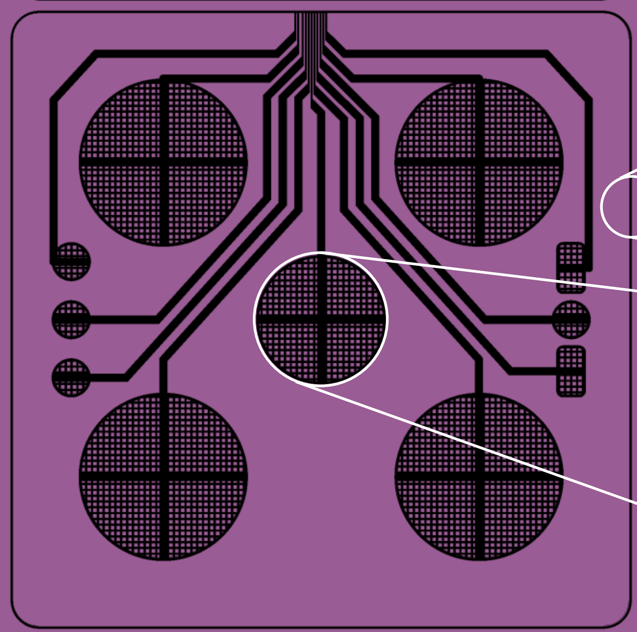
In order to create the pattern of holes in the mesh required to print the desired circuit, a mask of the **positive** of the circuit is needed first. This mask is created by inkjet printing the circuit design onto a transparent film (such as PET).

The positive mask is then placed on a photosensitive emulsion layer, or resist, which itself is placed on top of the fine mesh screen.
Exposing the photosensitive resist to light hardens the emulsion not covered by the mask while the covered emulsion stays soft. Washing off the covered soft emulsion leaves a **negative** of the circuit - a stencil - there are gaps in the resist where the ink can be squeegeed through.

The final negative mesh stencil might look like the image above.





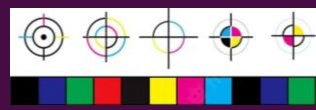


Printing Artefacts

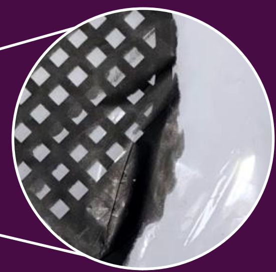
Mistakes and small details on the printed circuit layers give deeper clues about the manufacturing process and how the mat was assembled.

Ink test icons

A small crosshair symbol is printed at the edge of both layers. This is a symbol found on many bulk-produced printed items [6] to visually indicate ink level so that workers can see when the ink is running out.

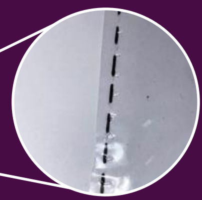


Ink seepage



The edges of some tracks are obviously smudged. This could be due to a small gap between the substrate (PET) and the mesh, allowing ink to ooze away from the gaps in the mesh to areas under the resist.

Cut/Sew lines



Around the edge of the sheet in some places a dashed line can be found.
This is likely to show factory workers where the sheet should be cut to size. It could also be a sewing line to guide the worker assembling the product.

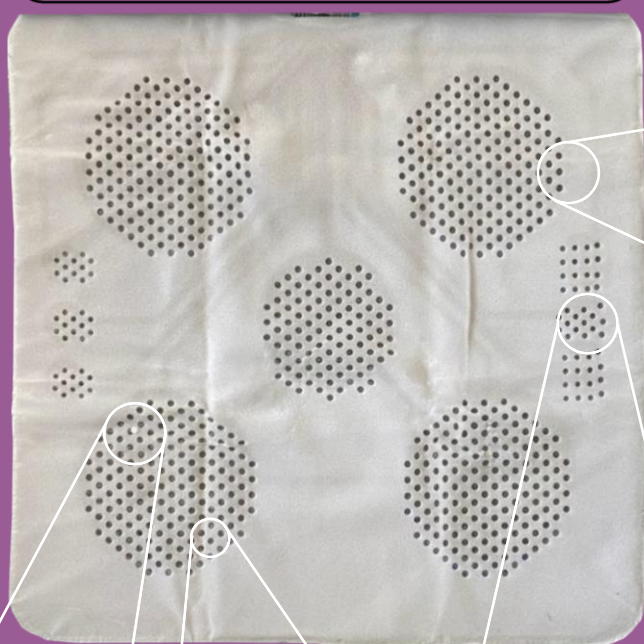
Defunct switch

There are graphite pads and foam holes for a centre switch, but the tracks lead to a pad on the PCB marked "**non-functional**" with no traces leading from it.

One possibility is that the designers opted not to use this functionality after the screen printing resists and foam cutting dies had been manufactured. It would be cheaper and quicker to edit the PCB than remake the tooling!

Translation:
"Centre (non-functional)"

Foam spacer layer



Material selection

Low-density, open cell polymer foam is used in between the circuit layers. This provides the resistance to switches being pressed and must be designed to ensure enough force is required to press the switch but not so much that it is hard for a child to press it.

As demonstrated, the use of this foam is successful here (3kg activation force). The foam also makes the mat softer to stand on and adds some friction between the very slippery PET circuit layers, so the user doesn't slip and fall over!

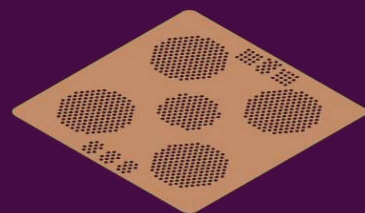
As with the other materials in the mat, the foam is cheap, lightweight and thin – allowing cheap, fast production.

Die cutting

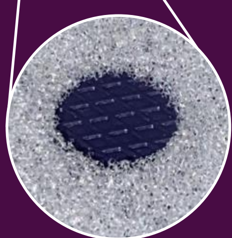
A common method for cutting foam sheets into set patterns is die cutting. Like a very large cookie cutter, this involves pressing the die into the foam (often hydraulically [7]) to cut out the material.

The die can be cheaply made by laser cutting slots into a piece of wood, where steel cutting blades can be fixed to create the desired pattern.
Custom die cut EPE foam bought in bulk on Alibaba is available for less than £0.40/sheet.

However, the downside of this process is that the die cannot be changed once made. If a mistake is found and needs to be reworked, then production must stop, and a new die must be made.

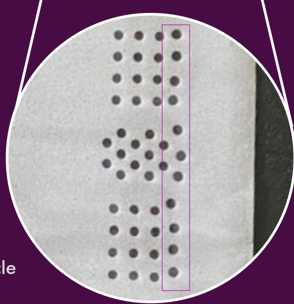


Castaway



Deconstructing the dance mat revealed a single cut out circle of foam leftover from the die cutting process. Cleanly cut compressed edges and perfectly regular hole sizes and spacing are tell-tale signs of die cut foam.

Reworked holes



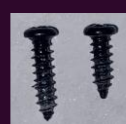
Inspecting the right-hand side switch holes reveals that the edge column is misaligned and larger than the others. One possibility is that this is an in-house rework of the part, where a worker has quickly stamped out (by hand) an extra column of holes on each foam layer.

There are many reasons this may have been required. For example, if all the foam had already been cut and then the designers realised that the holes and switches were misaligned it would save money and time to rework the existing cut foam instead of having more cut.

Improvements

The construction of the dance mat is simple; assembly is quick, and all of the components are cheap and readily available. However, there are a few small tweaks to the design which might improve the design without significantly impacting cost.

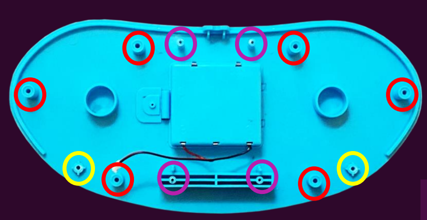
Standardise screws



6 M3 x 8 self tapping screws are used to fix the two halves of the controller housing together, while 4 M3 x 5 are used to fix the PCB into the lower housing piece and 2 M4 x 5 fix the mat to the controller.

If these screws are mixed up, then the PCB can become loosely fitting and rattles inside the housing – potentially damaging components – or the 2 halves of the housing do not fit tightly together.
The length of the bosses within the ABS lower housing means there is no need to have 2 slightly different screw lengths as either screw could work anywhere.

During reassembly time was wasted trying to work out which screw was needed where due to the visual similarity – standardising the screw lengths would eliminate this and speed up assembly.

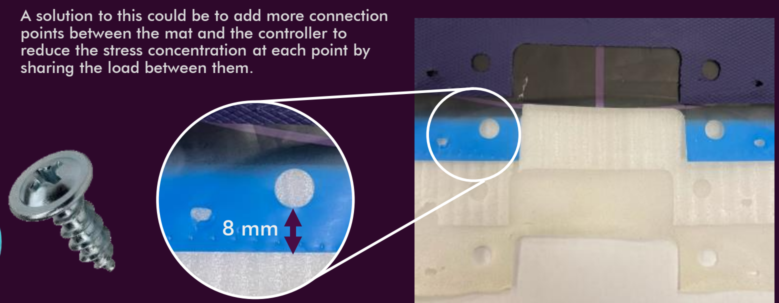


Stronger Mat-Controller interface

The mat is fixed to the controller with 2 x M4 self-tapping screws, and around 2 8mm bosses. This creates a stress concentration point in the mat; only 8 mm of material separates the edge of the mat from the hole. Pulling on the mat from the housing repeatedly may risk tearing this hole.

A strong connection between the mat and controller is needed because the flexible printed circuits directly connect to the PCB. Relative movement between the mat and controller could strain the fragile PET film which breaks easily – causing irreparable damage to the mat.

A solution to this could be to add more connection points between the mat and the controller to reduce the stress concentration at each point by sharing the load between them.

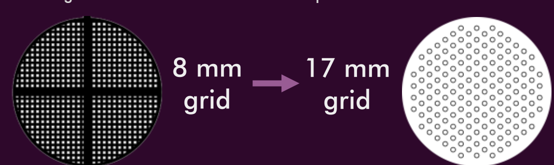


Optimising ink usage

The printed circuit layers have grids of lines in the activation area for each switch. The switches are activated by contacting the 2 circuits through the holes in the spacer layer.

The holes in the spacer layer are arranged in grids with rows and columns separated by **17 mm**, whilst the graphite grid on the switch layer contact pads has rows separated by **8 mm**. This has likely been designed as a 1:2 ratio to doubly ensure that there is graphite under every hole - so that every step is registered.

A cost improvement could be to align the sizes of the grid of holes with the grid of printed graphite – reducing the amount of conductive ink required.



The graphite grid in the contact pads could be reduced to 17 mm spacing, however this may be problems with alignment. To combat this the foam spacer layer could be laser cut instead of die cut.

Laser cutting is more accurate and would reduce alignment issues. It would also avoid any messy reworking shown currently in the product.

References: 1. ABS cost/kg from: plasticker.de; 2. Bulk sheet material, electronics and fasteners cost estimations from: alibaba.com; 3. Pool noodle photo from: amazon.co.uk; 4. Foam die example photo from: truformlaserdies.com; 5. Embossing roller photo from: made-in-china.com; 6. CMYK Ink test icons photo from: freepik.com; 7. Hydraulic die press photo from: hw-thermoforming.com